

Draft - expert review pending

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Topic 3.1 - From patterns to algebra: variables and expressions

A real classroom situation

One triangle, then more: the pattern and its rule



1 triangle
3 sticks



2 triangles
5 sticks



3 triangles
7 sticks



4 triangles
9 sticks

The rule

$$\text{sticks} = 2 \times \text{triangles} + 1$$

each new triangle shares one side,
so it adds only 2 sticks

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During an activity period in a Kotri school, pupils lay matchsticks in triangles. One triangle takes 3 sticks. Two triangles joined in a row take 5. Three take 7. A pupil, Amna, suddenly says: "Every new triangle adds only 2 sticks, because"

shares one side." The teacher asks: "Then how many sticks for 20 triangles?" Amna thinks and says: "3, then 2 more each time, so 3 plus 38 more... 41." She is right. But her method recounts from the start every time. Algebra's gift is shorter: name the unknown (t triangles), find the rule (sticks = $2 \times t + 1$), and answer for any t at once: for $t = 20$, sticks = 41. This topic is the road from Amna's method to the rule.

Explanation

Patterns and sequences

A pattern is a rule that repeats or grows, and a sequence is its list of outputs: 3, 5, 7, 9, ... for the triangles. The heart of pattern work is finding **how the pattern grows**:

- **Growing by a fixed step** (add 2 each time): 3, 5, 7, 9, 11, ...
- **Growing by doubling**: 2, 4, 8, 16, ...
- **Repeating**: 1, 2, 3, 1, 2, 3, ...

The matchstick pattern grows by a fixed step of 2, which is why its rule is of the form "2 times something plus a constant". Spotting the growth type is the first reasoning move; the rule follows from it. Gula and Lovric (2025) call this kind of crossing the transfer between concrete and abstract thinking spaces: the sticks on the desk are the concrete, the rule $2t + 1$ is the abstract, and a quality numeracy task makes the learner cross that bridge in both directions. The national mathematics curriculum for Grades 1-12 (NCC, n.d.) spans exactly the years in which pupils first meet patterns and their rules; this topic walks the same road a teacher must walk with them.

Variables and algebraic expressions



Term	What it means	Classroom example
Variable	a letter for a quantity that can change	the number of triangles
Coefficient	the number multiplying the variable	the 2 in $2t + 1$; 30 in $500 + 30k$
Term	a part of an expression, split by + or -	2t and 1, the terms of $2t + 1$
Expression	a rule from variables and numbers; nothing is claimed equal	$2t + 1$, the matchstick rule; $500 + 30k$, the van cost rule
Equation	a claim that two expressions are equal for some value of the variable	$2t + 1 = 41$ asks: how many triangles need 41 sticks?

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A variable is a named quantity that can change: t for the number of triangles, r for kilometres travelled, m for monthly fee rupees. The word "variable" is honest - it does not mean "unknown mystery"; it means "this quantity varies, and the rule holds for every value it takes". An algebraic expression is a rule written with variables and numbers: $2t + 1$ is the matchstick rule. The table above fixes the five words you must use precisely from now on: variable, coefficient, term, expression, equation. In $2t + 1$, the term $2t$ carries the coefficient 2, and the whole rule is an expression; the moment we write $2t + 1 = 41$ it becomes an equation, because now something is claimed to equal something.

Worked example. A school van charges 500 rupees to hire plus 30 rupees per kilometre. Write the cost rule. The quantities: a fixed 500 and a variable distance k . The expression: cost = $500 + 30k$. For a 12-kilometre trip: $500 + 30 \times 12 = 860$ rupees. Notice the expression was built from the situation's structure (fixed part plus per-kilometre part), not guessed.

A common misconception to correct. Trainees often read a letter as "a thing to solve for" and feel stuck when nothing can be solved: "What is t ?" The answer is: t is whatever the situation brings - 4 triangles, 20 triangles. The expression $2t + 1$ is not a puzzle; it is a machine that turns any t into its stick count. Reading expressions as machines, not riddles, is the shift this topic asks for.

Activity: Rule detective

Work in pairs. Your tutor gives each pair three growing patterns made from classroom materials (matchstick squares, dot arrays, tile borders). For each one: (1) build or draw stages 1 to 4, (2) record the sequence of counts, (3) state the growth step, and (4) write the rule as an expression using your own variable. Then swap with another pair and check their rules against stages 5 and 6, which they have not built. About 25 minutes.

Check your understanding

1. Write the first five terms of the sequence that starts at 7 and grows by a step of 4. (*Apply*)
2. For the matchstick pattern (rule $2t + 1$), compute the sticks needed for 12 triangles. (*Apply*)
3. In the expression $500 + 30k$, name the variable, the coefficient, and the two terms, and state what each represents in the van situation. (*Understand*)
4. A shop sells juice glasses at 45 rupees each plus a fixed box charge of 20 rupees. Write the cost expression and compute the cost of 8 glasses. (*Apply*)
5. A friend says " $2t + 1$ is unsolvable because we don't know t ". Explain in two sentences why the expression is a rule, not a riddle. (*Understand*)

Summary

Patterns grow by rules - by fixed steps, by doubling, or by repeating - and the growth type points to the rule's shape. A variable names a quantity that varies; an expression is the rule written with variables, a machine that turns each input into its output. The vocabulary (variable, coefficient, term, expression, equation) keeps the ideas separate, and reading expressions as machines rather than riddles is the shift that makes algebra usable.

Self-assessment checklist

Sign in to sync your ticks across devices.



- ☐ I can find the growth step of a pattern and predict later terms.
- ☐ I can write the rule of a growing pattern as an expression with a named variable.
- ☐ I can name the parts of an expression (variable, coefficient, term) in a daily-life situation.

Try this at your practicum school

Find one repeating or growing arrangement in the school (border tiles around a notice board, desks in rows, flags on a string). Record its first four stages, find the growth step, and write its rule as an expression. Check the rule against a fifth stage you count directly.

Summative task

A decorator hangs bunting in triangles along the school gate for a function. One triangle needs 3 metres of ribbon; each extra joined triangle adds 2 metres; and there is a fixed 5-metre tail at each end of the string. In a short written answer: (a) write the ribbon expression for n triangles, (b) compute the ribbon for 15 triangles, and (c) the school has 70 metres of ribbon: state the largest whole number of triangles the string can carry, and show the check.

Mini-rubric (each criterion 0-2 marks):

- *Expression*: $2n + 1$ for the triangles plus 10 for the two tails, i.e. $2n + 11$ (either built up or stated with parts labelled).
- *Computation*: $2(15) + 11 = 41$ metres.
- *Largest n with check*: $2n + 11 \leq 70$ gives $n \leq 29.5$, so 29 triangles, with the substitution check $2(29) + 11 = 69 \leq 70$ shown. **(Analyze)**

Further reading

- Marecek, L., Anthony-Smith, M., & Mathis, M. H. (2020). *Prealgebra 2e* (Ch. 2 The Language of Algebra). OpenStax.
<https://openstax.org/details/books/prealgebra-2e>



- Gula, T., & Lovric, M. (2025). Promoting mathematical thinking of university students: The case of a numeracy course. *Canadian Journal of Science, Mathematics and Technology Education*, 25(1), 171-184.
<https://doi.org/10.1007/s42330-024-00342-0>

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